

Do LLM-Generated Resumes Make Me More Qualified? An Observational Study of LLMs For Resume Generation and Matching Tasks

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Abstract—Large Language Models (LLMs) are gaining traction in various hiring-related tasks for both candidates and employers. For instance, employers are increasingly using LLMs to rate applicants based on the match between their resume content and job description requirements. Meanwhile, candidates use LLMs to write cover letters and tailor resume content. However, research shows LLMs can impart biases for LLM-generated content and against minorities, such as disabled candidates in hiring contexts. Thus, we conduct an observational study to investigate how widely used LLMs – OpenAI’s GPT-4, Google’s Gemini and Anthropic’s Claude – perform in supporting resume tasks for employers and candidates. Using a real-world dataset of job descriptions and resumes from disabled ($n = 209$) and non-disabled ($n = 209$) candidates, we examine the capabilities of these models across resume rating and resume generation tasks in a zero-shot setting. Our main findings show moderate alignment across models for resume matching and no significant differences in ratings between disabled and non-disabled candidates. However, we did observe increased ratings for LLM-generated resumes for GPT-4 and Claude. We discuss the implications for both candidates and employers based on our findings, aiming to promote non-biased and equitable AI-based hiring processes and motivate human-AI collaboration in the design of future hiring systems.

Index Terms—Resume rating, Resume generation, LLM hiring bias, LLM resume generation

I. INTRODUCTION

One of the first stages in hiring pipelines is for candidates to create and submit resumes—or documents outlining candidates’ qualifications and experiences [40]—to apply for open positions. Then, employers parse and rate submitted resumes across various hireability-based dimensions (i.e., work experience, skills, education, and certifications [60]) to identify top candidates matching the job requirements to advance to the subsequent hiring stage. Thus, *resume generation* and *resume matching* are critical for determining employment outcomes and hiring decisions—enabling the selection of highly qualified candidates. However, resume-related tasks in hiring pipelines are difficult. For employers, resume matching is becoming more complex as companies are often overwhelmed by the sheer number of applications submitted to open positions [58]. For candidates, resume generation is challenging as candidates face the stiff task of summarizing their qualifications in a maximum of two pages, as preferred by recruiters [13], [22].

To streamline hiring, artificial intelligence (AI) is becoming more integrated in hiring pipelines. For instance, most Fortune 500 companies use Applicant Tracking Systems (ATS), which are frequently powered by AI [34], to accept, parse, and rank resumes before human recruiters review them [59]. Recently, Large Language Models (LLMs) have shown promise in supporting hiring tasks—such as video interviewing [33] and technical [18] and personality assessments [12] for employers, while candidates use them to write cover letters [14] and prepare for job interviews [6]. The textual context understanding abilities of LLMs—capable of understanding and inferring information from human language [67]—also make them a better fit for resume-related tasks compared to traditional keyword-based approaches [38]. As such, LLMs are increasingly used to enhance hiring processes that rely on resumes. To make their resumes stand out, candidates use LLMs to generate the first drafts of resume documents [55], tailor the content of resumes [50], and refine content according to job descriptions [68]. For employers, LLMs are increasingly used for resume matching [36], and have been shown to save manual effort and time for recruiters [26]. Thus, LLM-based *resume generation* and *resume matching* are poised to become the new normal in hiring contexts.

While previous research explores the capabilities of LLMs in resume generation and matching, there are LLM-specific concerns that could have negative effects in hiring contexts. For example, *self-preference*, or the extent to which LLMs favor LLM-generated text over non-LLM-generated text [47], [64], can incorporate potential bias for LLM-generated resumes while disadvantaging human-written resumes. In addition, there are concerns related to diversity, equity, and inclusion (DEI), such as *algorithmic bias* against minorities in AI-driven hiring [66] and LLM-based resume matching [7]. For example, Glazko et al. show that GPT-based resume screenings impart bias against candidates with disabilities [29]. In particular, *resume whitening*, or removing DEI-related signals from resumes [27], may persist in LLM-generated resumes to minimize bias, harming users’ self-identity [2]. Additionally, many recent works propose multi-agent systems in hiring contexts [26], [38], [49], which pose a greater risk of extrapolating AI concerns.

To address these concerns, it is important to understand

LLMs in resume matching and generation tasks, regarding their alignment across model ratings, DEI dimensions, and rankings of human and LLM-written resumes, to motivate future system designs. For this preliminary study, we focus on disability, as disability is often ignored in DEI initiatives [8], including aspects of diversity on websites for Fortune 500 companies [39]. Further, individuals with disabilities are increasingly using LLMs for various tasks, including writing resumes and thank-you letters to employers [44]. To the best of our knowledge, we are the first to study self-preference and resume whitening in LLM-driven resume tasks. Through our work, we aim to tackle following research questions (RQs):

RQ1: How aligned are LLM-generated resume ratings across candidates' ability and resume rating constructs? **Motivation:** Understanding resume rating alignment across popular LLMs in zero-shot settings will provide initial insights into their rating heuristics and compatibility for future multi-agent systems.

RQ2: How do LLM-generated resumes fair for disabled candidates with regards to resume whitening? **Motivation:** Understanding resume whitening is important, as the loss of critical information can impact resume ratings and lead to bias against disabled candidates in resume matching.

RQ3: How do resume ratings for human-written resumes compare to LLM-generated resumes? **Motivation:** Insights into LLM self-preference will help minimize potential biases against human-written resumes in future LLM-based resume matching systems.

We answer the research questions in an observational study, exploring the capabilities of three popular LLMs—GPT-4, Claude, and Gemini—in generating and ranking the resumes of ($n = 209$) disabled and ($n = 209$) non-disabled real-world candidate resumes submitted across ($n = 24$) job titles. Our initial results show: a) moderate alignment across LLMs for work experience, skills and education; b) Claude exhibits traces of resume whitening for skills; and c) GPT-4 and Claude exhibit rating increases for LLM-generated resumes compared to human-written resumes. Based on our findings, we provide implications for enhancing the future of work with AI—informing the design of human-centric AI-based tools to promote human-AI collaboration for resume generation and matching tasks in hiring contexts.

II. BACKGROUND AND RELATED WORK

A. Resume Whitening

Prior work highlights the biased nature of job markets against minorities across diversity axes such as race [66], gender [41] and disability status [29]. One source of bias can be derived from direct and indirect cues from candidate resumes—such as names and participation in minority-related social clubs [3], [4], [51]. Resume whitening is a strategy used by minorities to downplay diversity markers in their resumes [35] to minimize bias effects. For instance, Gerdeman et al. [27] found an increase of 15% in interview callback rates for Black candidates after whitening their resumes. Most resume

whitening strategies suggest markers that reveal candidates' gender or race/ethnicity should be removed from resumes to avoid AI-based bias [5]. While some candidates view this process as necessary, resume whitening can also have adverse effects on candidates [35]. For example, the loss of information can lead cause emotional and psychological damage [2] while perpetuating systemic bias in hiring processes [46].

B. Bias in Hiring

Prior work also outlines the challenges that candidates with disabilities face in hiring contexts [23]—noting biases in interview processes [54], career development [30], and workplaces [20]. Bias issue is also continued in the LLM era. LLMs are black-box in nature, hence prior work highlights numerous ethical considerations about the use of LLMs in hiring contexts. LLMs have shown gender bias against females in writing cover letters [62], generating interview responses [37], resume generation and resume assessment [9] and scenario-based tasks [31]. LLMs have also shown demographic bias [24] and discrimination based on names and marriage status [61]. An et al [7] and Gabler et al [25] show LLMs exhibit ethnic bias against marginalized populations. Prior work shows also demonstrate bias against candidates with disabilities in AI-based resume matching [29], [42], [56]. However, previous studies focused on disabled candidates in LLM-based hiring contexts are experimental [29], [63], holding tight control over factors that can influence LLM ratings and limiting the generalization of the findings. We concentrate on disability to understand strategies and potential biases in modern LLMs for generating and rating resumes.

C. Resume Generation

Resume generation refers to the process of candidates constructing resume documents—either manually or automatically—based on their experiences and background. There is limited prior work in terms of evaluating LLM performance in resume generation tasks. Most prior work in LLM-based resume generation aims to fine-tune resumes according to job descriptions. For example, Zijad et al. [68] propose ResumeFlow, a novel tool which outputs a tailor-made resume according to job description requirements. Similar work is conducted by Patra et al. [50], exploring Retrieval Augment Generation (RAG) to tailor resumes. While prior studies focus on tailoring resumes to job requirements, no prior work has studied LLM-based resume generation.

D. Resume Rating

Resume rating is the process of examining the extent to which resumes match the requirements outlined in job descriptions. In terms of resume rating, most prior work is based on resume matching [11], [21], [28] where resume is matched to a particular job description based on the resume content directly without any resume ratings. This fails to reflect ATS systems in real-world applications where each resume is rated on a more fine-grained level across different resume rating constructs such as work experience match,

skills match, education match, etc. following which candidates with the highest ratings are shortlisted. Vaishampayan et al [60] is the closest work with our work. However, [60] only considers GPT-4 resume ratings, comparing them with human ratings, and only explores potential bias related to gender and race/ethnicity. Our work is based on multiple LLMs – GPT-4, Claude and Gemini. In addition, we compare the generated and rated resumes across disabled and non-disabled candidates—motivated by prior work [29]. Additionally, we study LLM self-preference, providing a deeper understanding of LLM ratings for resumes generated by humans and models.

III. DATASET

Our industry partner (Blinded for Review) provided us with a real-world dataset of ($n = 35,138$) candidate resumes submitted across ($n = 79$) job titles. This partner is a small-scale recruiting firm with over 30 years of experience in the various domains. The dataset spans resumes from Jan 2022 - Mar 2023, therefore the probability of LLM-generated resumes in the dataset is low as modern LLMs were released after Nov 2022. The dataset was shared as a CSV file where the resumes and job titles were accompanied with the exact job description along with candidate reported demographic information such as gender, race and disability status. The dataset in total had ($n = 209$) disabled candidate resumes, based on self-reported disability status in the CSV file, submitted across ($n = 24$) job titles such as administrative assistant, office manager, sales, etc. We matched the distribution of disabled candidate resumes and sampled ($n = 209$) resumes from non-disabled candidates submitted to the same job titles using stratified sampling [48]. To minimize selection bias, we randomly made ($n = 10$) sets of samples for non-disabled candidates and picked one randomly as our final sample. Our final dataset consisted ($n = 209$) each disabled and non-disabled candidate resumes submitted to ($n = 24$) jobs. Table I shows the top-10 job titles with most number of resumes. Due to the Institutional Review Board (IRB) restrictions, we are not able share the original dataset in our supplementary materials as the dataset contains personal information.

TABLE I: Top-10 Job Titles for Disabled Candidate Sample According to No of Resume

Job Title	No of Resumes
Accountant and Finance	36
Administrative Assistant	27
Recruiter	24
Sales	15
Marketing	15
Office Manager	13
Project Manager	13
Receptionist	9
Project Coordinator	9
Supervisor	6

IV. METHODOLOGY

To answer our research questions, we observed the capabilities of LLMs in resume generation and resume rating tasks.

A. Resume Rating

We followed the prompt structure used by Vaishampayan et al [60] in their resume rating paper. We reused the same prompt across GPT-4, Claude and Gemini. We prompted the LLMs through API calls at zero-shot setting and using the latest versions of all three LLMs. We ran a pilot study with different prompts, temperature values and system roles, and manually inspected responses to verify the respective settings used for API calls gave the best response. We prompted GPT-4o, Claude Sonnet-3.7 and Gemini Flash-2.0 with a temperature value of 0.2, providing consistency in LLM responses and minimizing hallucinations. The role of the system was set to “professional HR that rates resumes”. The prompt used for resume rating had the following structure:

Generate a score on the scale 1-5 for each work experience match, skills match, educational background match and certifications/extracurricular match based on the job description summary and resume. Additionally provide the reasons for the generated rating. Be strict in rating. [job description summary] [resume content]"

B. Resume Generation

We followed same processing used in resume rating for the resume generation task. The only difference was for system role which was set to “professional HR that creates resumes”. The prompt used in resume generation had the following structure:

Generate the resume in text format based on the provided content [resume content]"

C. Data Analysis

RQ1 analyzes the alignment, or agreement, in resume ratings between all three LLMs. Fleiss’ kappa [45] measures the inter-rater agreement between multiple raters, and is widely used in literature. For our case, Fleiss’ kappa shows the level of agreement between LLM ratings. Higher agreement demonstrates higher exact LLM rating matches. For our study, *Fleiss kappa*- values of 0–0.19 indicate “poor”, 0.20–0.40 is “fair”, 0.41–0.60 is “moderate”, 0.61–0.80 is “substantial” and above 0.8 is “almost perfect” agreement respectively. While Fleiss’ kappa interprets the exact rating matches between LLMs, it fails in capturing the association between LLM ratings. We measure association through Pearson’s correlation [52], complementing the Fleiss’ kappa value. For our study, *Correlation*-values of 0–0.19 indicate “very weak”, 0.20–0.40 is “weak”, 0.41–0.60 is “moderate”, 0.61–0.80 is “strong” and above 0.8 is “very strong” association respectively. High correlation along with high agreement demonstrates higher LLM rating alignment. Additionally, to explore the differences in rating schemes followed by the specific LLMs, we undertook open-coding based thematic analysis. RQ2 explores potential bias in LLM-based resume tasks for candidates with disabilities. To analyze resume whitening, the first author manually inspected

LLM-generated and actual disabled candidate resumes, specifically observing the removal of disability markers in generated resumes. Additionally, we analyze the disabled and non-disabled resume ratings between both the subgroups. RQ3 compares the resume ratings of LLM-generated and human-generated resumes. For analyzing RQ3, we use statistical analysis, via a one-tailed t-test, to compare resume ratings of LLM-generated and non-LLM-generated resumes.

V. RESULTS

In this section we present our results to our research questions.

A. RQ1: Alignment of LLMs

We measured alignment in resume ratings for disabled and non-disabled candidates by comparing the generated resume ratings for GPT-4, Claude and Gemini across the dimensions of work experience, skills, education and certification. We report two measures, a) Pearson’s correlation and b) Fleiss’ kappa. The CSV file containing all resume ratings can be found in our supplemental materials.

TABLE II: Correlation Matrix for Work Experience and Skills (Disabled Candidate Resumes)

	Work Experience			Skills		
	GPT	Claude	Gemini	GPT	Claude	Gemini
GPT	1.000	0.812	0.800	1.000	0.700	0.668
Claude	0.812	1.000	0.796	0.700	1.000	0.748
Gemini	0.800	0.796	1.000	0.668	0.748	1.000

TABLE III: Correlation Matrices for Education and Certification (Disabled Candidate Resumes)

	Education			Certification		
	GPT	Claude	Gemini	GPT	Claude	Gemini
GPT	1.000	0.729	0.669	1.000	0.622	0.468
Claude	0.729	1.000	0.629	0.622	1.000	0.553
Gemini	0.669	0.629	1.000	0.468	0.553	1.000

TABLE IV: Fleiss’ Kappa Scores Between LLMs (Disabled Candidate Resumes)

LLM Pair	Work Exp	Skills	Education	Cert
GPT, Claude, Gemini	0.4967	0.3302	0.3355	0.2278
GPT and Claude	0.5088	0.4312	0.4227	0.2686
GPT and Gemini	0.5185	0.2010	0.3275	0.1776
Claude and Gemini	0.4607	0.3508	0.2554	0.2064

1) Candidates’ Ability:

a) *Disabled candidate resumes:* For disabled candidate resumes, we found “very strong agreement” across work experience and “strong agreement” across skills and education for all three LLMs. Certifications saw the least agreement between LLMs, where GPT-Claude had “strong agreement” but GPT-Gemini only had “moderate agreement”. Tables II and III provide the correlation metrics for all three LLMs. Similar results were reflected for Fleiss’ kappa, where we observed “moderate” agreement for work experience, skills

and education but “fair” agreement for certifications. Overall, GPT-Claude saw the highest agreement whereas GPT-Gemini had the least agreement. Table IV provides in-depth details.

b) *Non-disabled candidate resumes:* All LLM pairs exhibited “strong” correlation across the dimensions of work experience, skills, education and certifications. But interestingly, this was not reflected in Fleiss’ kappa where we observed “poor” agreement for certification across GPT-Gemini and Claude-Gemini pairs. Specifically, strong correlation highlights the fact that the ratings for all pairwise LLMs increase or decrease together, but “poor” Fleiss’ kappa agreement underlines misalignment between the resume ratings in most of the cases. These results are presented in Tables V, VI, and VII.

TABLE V: Correlation Matrix for Work Experience and Skills (Non-Disabled Candidate Resumes)

	Work Experience			Skills		
	GPT	Claude	Gemini	GPT	Claude	Gemini
GPT	1.000	0.774	0.780	1.000	0.658	0.707
Claude	0.774	1.000	0.759	0.658	1.000	0.660
Gemini	0.780	0.759	1.000	0.707	0.660	1.000

TABLE VI: Correlation Matrices for Education and Certification (Non-Disabled Candidate Resumes)

	Education			Certification		
	GPT	Claude	Gemini	GPT	Claude	Gemini
GPT	1.000	0.704	0.706	1.000	0.690	0.668
Claude	0.704	1.000	0.713	0.690	1.000	0.651
Gemini	0.706	0.713	1.000	0.668	0.651	1.000

TABLE VII: Fleiss’ Kappa Scores Between LLMs (Non-Disabled Candidate Resumes)

LLM Pair	Work Exp	Skills	Education	Cert
GPT, Claude, Gemini	0.4205	0.3528	0.3524	0.2161
GPT and Claude	0.4451	0.4305	0.4145	0.3265
GPT and Gemini	0.4510	0.3109	0.2718	0.1690
Claude and Gemini	0.3605	0.3092	0.3701	0.1180

To gain a deeper understanding of misalignment between LLMs across rating constructs, we sampled all the instances from disabled and non-disabled candidates where the difference between any pair of LLMs ratings was substantial i.e. ≥ 2 . The first author manually inspected the LLM-provided rating reasons for each rating dimension separately, providing the following observations:

2) Resume Rating Constructs:

a) *Work Experience:* Work experience had least substantial differences ($n = 5$) across LLMs. We observed Gemini was more stringent in rating and was closely following the job description requirements in relevant work experiences, whereas Claude ratings were more lenient based more on assumptions for work experience in a similar field. For instance, for an interior design coordinator role the Gemini reasoning stated, “The candidate has extensive experience in the construction industry, but primarily in estimating and project management roles. The job description is for a Sales

Design Coordinator, which is more focused on design coordination and customer service. While the candidate's experience in contract management and material procurement could be somewhat relevant, the core responsibilities are different". But Claude responded, "The candidate has extensive experience (40 years) in the construction industry, far exceeding the required 3-5 years.... However, the rating is not a perfect 5 because their specific experience is more focused on tile, marble, and HVAC contracting rather than interior design coordination as specified in the job description". Here, Gemini follows the requirement closely and provides a low rating to the candidate whereas Claude considers 40 years of experience in similar field as relevant, even after stating that candidate does not actually have exact relevant work experience.

b) Skills: We recorded ($n = 12$) instances of substantial disagreement between LLMs. On closer inspection, we observed an exact opposite case to work experience where we observed Gemini was most lenient and hallucinated in ($n = 5$) cases overstating the skills of the candidate where as GPT ratings were the most stringent closely matching the required skills with the candidate resume contents. Even missing out on one skill for GPT would receive a lower ratings of 1 or 2 out of 5. For example, for a Logistics coordinator role, the resume of a candidate did not demonstrate knowledge of Federal Law 21CFR820, and therefore GPT reasoned, "The resume does not demonstrate specific knowledge of quality regulations or Federal Law 21CFR820, which are skills required for the job. This results in a skills match rating of 2". On the other hand, since the candidate had a few related skills, Gemini gave a rating of 4 along with the reasoning, "The candidate demonstrates relevant skills such as logistics, supply chain management, procurement, Hazmat handling, and quality assurance. However, there is no mention of knowledge of quality regulations or Federal Law 21CFR820".

c) Education: Education recorded ($n = 34$) substantial disagreements between the LLMs. Most of these ($n = 20$) were between GPT-4 and Gemini. Closer inspection revealed rating scale differences to be responsible for these disagreements. In cases where the job description did not specify any minimum educational requirements and candidate resumes did not contain information regarding their educational background, GPT gave a minimum rating of 3. However, Gemini gave a rating of 1, with the reasoning that educational qualifications were missing in the resume. On the other hand, for cases where candidate resumes met the minimum required qualification as per the job description, GPT provided a rating of 3 but Gemini gave a rating of 5, as candidate met the minimum requirements. Therefore, there is misalignment between GPT-4 and Gemini rating scales for education, which resulted in disagreement.

d) Certification: Certifications had the highest ($n = 51$) misalignment. Similar to education, most of these disagreements were due to rating scale differences as most ($n = 42$) job descriptions did not specify any minimum certification requirements, leading to disagreements between GPT-4 and Gemini. The final summary of RQ1 findings is provided in

Figure 1.

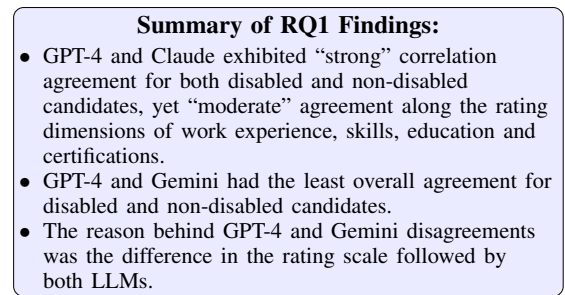


Fig. 1: Summary of Key Findings in RQ1

B. RQ2: Resume Whitening

To analyze whether LLMs employ resume whitening strategies, we first selected a sample of resumes based on the self-reported disability status of candidates (see Section III). After manual inspection, we found ($n = 37$) resumes contained direct or indirect disability markers, such education in special colleges, medical leave in work experiences, medical discharge due to disability, etc. The LLM-generated resumes for these samples were inspected again by first author to specifically observe the presence of detected disability markers. Our inspection found the following observations:

a) GPT & Gemini did not whiten resumes: Across ($n = 35$) resumes, we found disability markers were present in GPT and Gemini-generated resumes. There were ($n = 2$) and ($n = 1$) instances where disability markers were absent in GPT and Gemini-generated resumes, respectively. However, those were due to the length of the original resume content and limited number of output tokens for LLMs to generate the resumes. When we reduced the length of the original resume, disability markers appeared in the GPT and Gemini-generated resumes.

b) Claude showed traces of resume whitening: We found six instances of resume whitening from Claude. For ($n = 2$) instances, disability markers across skills were removed, such as dementia care ($n = 1$). There was also an instance from work experience where a candidate declared taking a medical break, but this statement was missing in the Claude-generated resume. To re-affirm this observation was not by chance, we made ($n = 20$) copies of the resumes where skills-based disability markers were missing and manipulated them in following ways: a) putting the disability markers at the beginning of skills section; b) duplicating the disability markers multiple times; c) putting the disability marker as the only skill; and d) changing the name of the candidate. Out of ($n = 20$) instances, there was only one instance when the Claude-generated resume contained the disability marker—when it was listed as the only skill in the manipulated resume. The remaining instances ($n = 19$, 95%) had the disability marker removed. For the medical break-related resume, we followed the same process as above, making ($n = 20$) copies of the resume manipulated in the following ways: a) putting the sentence regarding the medical break in summary section at the beginning of the resume; b) putting it at the top in the work

experiences section; c) duplicating it multiple times in different sections; and d) changing the name of the candidate. In all ($n = 20$) instances, we found the medical break statement was missing from the Claude-generated resume. The exact csv files containing the disability markers and manipulations can be found in our supplementary materials. The final summary of RQ2 is provided in Figure 2.

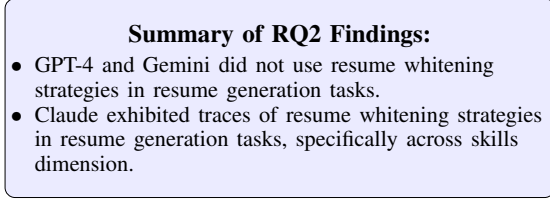


Fig. 2: Summary of Key Findings in RQ2

C. RQ3: Self-Preference

We explore the self-preference trait of LLMs in resume rating contexts through RQ3. To compare ratings for LLM-generated and human-generated resumes, we first formulated the resume ratings by prompting GPT-4, Gemini and Claude across the dimensions of work experience, skills, education and certifications. Then, we pass the human-written resume as text into LLMs and prompt the models to generate a resume based on the same human-written resume text. Finally, we prompt the LLMs again to generate resume ratings based on the LLM-generated resume in the previous step. In an ideal scenario, since both resumes are based on the exact same content they should have the same ratings. For statistical confirmation, we use the one tailed t-test to check if the ideal condition holds true.

a) *Disabled Candidate Resumes:* We found that, on average, GPT and Claude-generated resume ratings were higher across work experience, skills, education and certifications than the respective ratings for human-generated resumes. For Gemini, we saw mixed results. There was an increase across work experience and skills, but reduction in ratings for certifications. T-test results show that the increase rating for skills in Claude-generated resumes was statistically significant when compared to human-generated resume ratings. Table VIII provides detailed t-test results and average ratings.

TABLE VIII: T-test Results and Average Ratings (scale of 1-5) for LLM-Generated vs Human Resumes

Model	Aspect	Human Avg	LLM Avg	Diff	p-value
GPT-4	Work Exp	2.98	3.01	0.03	0.365
GPT-4	Skills	3.13	3.22	0.09	0.119
GPT-4	Education	2.91	2.92	0.01	0.463
GPT-4	Cert	2.53	2.55	0.02	0.244
Claude	Work Exp	3.14	3.29	0.15	0.069
Claude	Skills	3.21	3.39	0.18	0.012
Claude	Education	2.88	2.97	0.09	0.201
Claude	Cert	2.25	2.38	0.13	0.057
Gemini	Work Exp	2.92	2.95	0.03	0.381
Gemini	Skills	3.61	3.63	0.02	0.367
Gemini	Education	2.63	2.65	0.02	0.448
Gemini	Cert	1.96	1.86	0.10	0.143

b) *Non-Disabled Candidate Resumes:* We observed a similar trend for non-disabled candidates, with an increase in ratings for GPT-4 and Claude across all rating dimensions. Similarly, Gemini-generated resume ratings for certifications dropped when compared to human-generated resumes. However, this decrease was not statistically significant. We found the increase in Claude-generated resume ratings across work experience and skills to be statistically significant when compared to human-generated resume ratings. Table IX provides the detailed results and figure 3 shows average ratings.

TABLE IX: T-test Results and Average Ratings for LLM-Generated vs Human Resumes (Non-Disabled Candidates)

Model	Aspect	Human Avg	LLM Avg	Diff	p-value
GPT-4	Work Exp	2.97	3.04	0.07	0.218
GPT-4	Skills	3.17	3.29	0.12	0.052
GPT-4	Education	3.03	3.03	0.00	0.463
GPT-4	Cert	2.53	2.57	0.04	0.336
Claude	Work Exp	3.18	3.28	0.10	0.045
Claude	Skills	3.26	3.39	0.13	0.034
Claude	Education	2.94	3.03	0.09	0.186
Claude	Cert	2.33	2.36	0.03	0.369
Gemini	Work Exp	2.89	2.95	0.06	0.283
Gemini	Skills	3.57	3.65	0.08	0.182
Gemini	Education	2.71	2.67	0.04	0.369
Gemini	Cert	1.85	1.79	0.06	0.298

To validate our findings, we randomly picked ($n = 25$) samples each from disabled and non-disabled resumes, totaling a set of ($n = 50$) samples. We made five such sets and repeated the same experiment again. Out of the five sets, we had to remove one (set 2) from data analysis as we found multiple instances of GPT-4 hallucinations. For the remaining four sets, our data analysis and t-test revealed similar results that further strengthens our findings. Detailed CSV files containing the five sets and the respective GPT-4, Claude and Gemini-based resumes along with ratings for LLM- and human-generated resumes can be found in our supplemental material.

TABLE X: T-Test Results for Human Resumes

Metric	LLM	t-value	p-value
Work Experience	GPT	-0.60186	0.2738
	Claude	-0.34484	0.3652
	Gemini	0.31183	0.3777
Skills	GPT	-0.61337	0.2700
	Claude	-0.72510	0.2344
	Gemini	0.34532	0.3650
Education	GPT	-1.11857	0.1320
	Claude	-0.50975	0.3052
	Gemini	-0.68502	0.2469
Certification	GPT	0.00000	0.5000
	Claude	-0.95971	0.1689
	Gemini	1.07676	0.1411

We also measured bias by comparing disabled and non-disabled candidate resume ratings across human and LLM-generated resumes. Using the one-tailed t-test, our results did not reveal any statistically significant differences in the ratings for disabled and non-disabled candidates, thereby failing to demonstrate bias. The results can be found in Tables X and

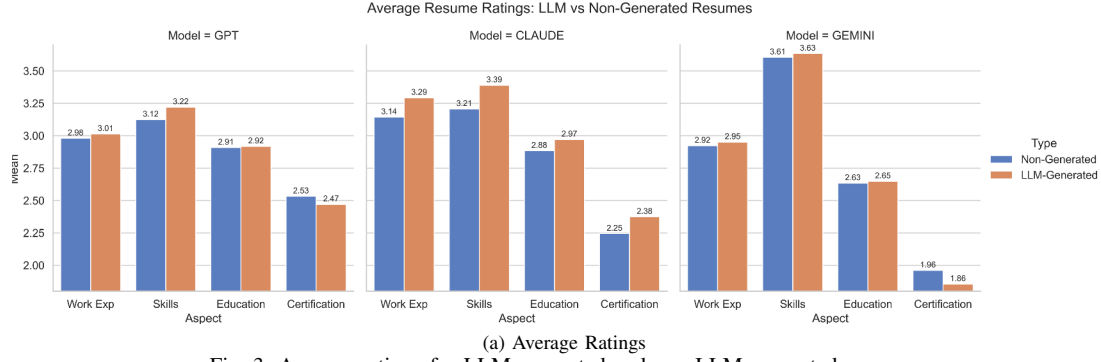


Fig. 3: Average ratings for LLM-generated and non-LLM-generated resumes

TABLE XI: T-Test Results for LLM-generated Resumes

Metric	LLM	t-value	p-value
Work Experience	GPT	-0.25499	0.3994
	Claude	0.14587	0.4420
	Gemini	0.05065	0.4798
Skills	GPT	-0.93719	0.1746
	Claude	-0.06197	0.4753
	Gemini	-0.22472	0.4112
Education	GPT	-1.12320	0.1310
	Claude	-0.56835	0.2851
	Gemini	-0.21170	0.4162
Certification	GPT	-1.11861	0.1320
	Claude	0.23177	0.4084
	Gemini	0.59581	0.2758

XI. The major summary findings for RQ3 are provided in Figure 4.

Summary of RQ3 Findings:

- GPT-4 and Claude exhibit self-preference, with increased ratings for LLM-generated resumes compared to human-written resumes across all rating dimensions. Gemini showed the least self-preference.
- Claude showed statistically significant increases in LLM-generated ratings across skills for disabled resumes and across work experience and skills for non-disabled resumes.
- No statistically significant difference between the ratings was observed for all three LLM resume ratings across human-written and LLM-generated resumes.

Fig. 4: Summary of Key Findings in RQ3

VI. DISCUSSION

Our findings suggest there is “moderate” alignment between LLMs across work experience, skills and education, whereas only “fair” alignment across certifications for resume ratings (refer to Section V-A). We also found Claude exhibits traces of resume whitening in resume generation, specifically across skills (refer to Section V-B). Finally, we found Claude exhibits the most self-preference, providing higher ratings for generated resumes over human-written resumes across all resume rating dimensions (refer to Section V-C). Based on

our findings, we discuss challenges for LLM-based *resume matching* and *resume generation* systems, and propose design guidelines to overcome these challenges in future systems.

A. Resume Matching

1) *Black-box Resume Matching*: Employers heavily use black-box deep learning-based systems [34], such as ATS [57] or Talent Management Software (i.e., Workday [1]), to automate resume matching tasks. While this automation saves time for recruiters [16], candidates have largely negative perceptions—distrusting black-box ATS systems [59] and demanding more clarity in terms of their decision making [10]. Decision making is based on resume ratings and thus can vary across different ATS systems. Future LLM-based systems will face similar challenge where decision making will depend on the choice of LLM. Our study underlines this challenge where we found all three popular modern LLMs – GPT-4, Claude and Gemini resume rating scales are different and therefore might lead to differing decision making.

Guideline: Provide an overview of rating processes to promote transparency Prior literature shows resume rating is based on constructs such as work experience, skills, etc. [19]. Therefore, employers could provide an overview of the rating processes for how candidates are evaluated to promote transparency in hiring processes. Further, this information can be incorporated into LLM-based rating systems to provide more aligned ratings and make them more open-box to candidates in terms of expectations and criterion used for decision making.

2) *Effective Resume Ratings*: Similar to resume parsing, accurate rating is critical. Keyword-based approaches rank candidates based exact keyword matches in resumes and job requirements, which can be inaccurate. For instance, JavaScript can also be written as “JS”, yet not accepting both options can negatively impact resume ratings [59]. LLM-based resume ratings can provide context understanding, but cannot fully replace human evaluations [60]. In our study, we observed GPT-4, Claude and Gemini rated skills leniently but were more stringent when rating certifications. This underlines challenges with LLM ratings in zero-shot settings, as modern LLMs can make associations between relevant skills and experience, yet fail to consider relevant certifications. For instance,

we saw an example where a PMP certification, which closely aligns with the construction field, was deemed irrelevant by GPT-4 and Gemini for a construction-related job. Thus, LLM-based resume ratings induce new challenges which need to be addressed to improve the effectiveness of resume ratings.

Guideline: Use multi-agent debate based resume rating

To overcome this challenge, multi-agent based systems can be used [32]. For instance, multi-agent debate-based systems can enhance decision making through multiple rounds of debate [15]. We envision a multi-agent feedback loop [53] can resolve the issues of rating scale differences and certification relevance, improving the overall effectiveness of resume rating.

3) *Fair Resume Ratings:* Prior work [41], [43], [56] underlines algorithmic biases against minorities in hiring contexts. While we did not observe statistically different LLM ratings across disabled and non-disabled candidate resumes and recent studies [60], [61] do not find bias against the gender and race of candidates, research shows LLM-based resume rating systems may also exhibit bias [7], [9]. Further, we observed self-preference for LLMs in resume rating tasks—which could lead to systems favoring LLM-generated or LLM-optimized resumes over human-written resumes. Thus, more work is required to investigate and minimize potential bias in LLM-based resume ratings.

Guideline: Use of Inference-time scaling strategies along with regular bias audits Regular bias audits can provide deeper insight into resume matching systems, thereby minimizing potential algorithmic bias against underrepresented candidates. Additionally, through inference-time scaling strategies, such as long chain-of-thoughts (CoT) [17], we can minimize self-preference bias of future LLM-based resume matching systems. Long chain-of-thoughts encourages the LLM to reassess their initial understandings and consider the alternative responses carefully. This makes LLMs think thoroughly before generating resume ratings, and thus will lead to LLM rating the resumes more fairly and consistently—potentially reducing self-preference biases.

B. Resume Generation

1) *The Double-Edged Sword of LLM-Based Resume Optimization:* Resumes are of utmost importance for candidates, as they have to first parse through ATS systems and before the slight chance of a human reviewing the resume [59]. To make their resume stand out, many candidates resort to using LLMs for tailoring the resume according to the specific job description requirements. While fine-tuning the resume through LLMs might be beneficial, it could be harmful. For instance, our study observed traces of resume whitening in LLM-generated resumes for Claude, specifically across work experience and skills. This loss of critical information in resume generation can negatively candidates [2] and resume matching processes [46].

Guideline: Human-AI collaboration is recommended for tailoring resume using LLMs Future resume generation systems could minimize resume whitening through HCAI-based Human-AI collaboration [65], where candidates are provided

the chance of selective tailoring – where they can select sections for AI-based generation and tailoring. Our study results show traces of resume whitening for Claude across work experience and skills dimensions. Thus, candidates can opt for skipping work experience and skills sections in order to avoid resume whitening when tailoring their resume using Claude.

VII. LIMITATIONS AND FUTURE WORK

We have a limited sample of ($n = 209$) disability candidate resumes submitted across limited job titles ($n = 24$). We also only investigate three LLMs, GPT-4, Claude, and Gemini. Therefore, our results might not generalize. We used stratified sampling to select a sample of non-disabled candidate resumes, but this might introduce selection bias. To minimize selection bias, we made 10 different sets of non-disabled resumes and randomly picked one (refer to Section III). Additionally, we only consider disability in terms of DEI and do not consider other aspects such as gender and race/ethnicity. We also conduct our study at zero-shot LLM settings, and our results might differ with fine-tuning or a few-shot learning.

Future efforts can extend our work to enhance generalizability. For instance, including open-source LLMs, such as Llama, instead of only commercial models. More in-depth analysis of responses can provide insights on reasoning of models to create the resume ratings. Future efforts can observe LLM rating alignment after a few-shot learning or fine-tuning. Our study only considers self-preference by comparing human-written and LLM-generated resumes, but future work can also study self-preference of LLM-LLM pipelines i.e., studying GPT-4 preferences for resumes generated by Claude. Finally, future work is needed to explore resume generation and rating tasks across other diversity dimensions.

VIII. CONCLUSION

We conducted an observational study for LLM-based resume rating and resume generation tasks, comparing disabled ($n = 209$) and non-disabled ($n = 209$) candidate resumes using three modern LLMs—GPT-4, Claude and Gemini. We evaluate disabled and non-disabled candidates’ resumes to study LLM alignment, resume whitening, and self-preference. Our findings show “moderate” alignment between GPT-4, Claude and Gemini for resume ratings across work experience, skills and education dimensions, yet “fair” alignment across certification. We also found Claude shows traces of resume whitening, specifically across skill dimension. Finally, self-preference is observed in both GPT-4 and Claude, where the latter showed a significant preference for LLM-generated resumes across skills and work experience. In light of our findings, we discuss challenges and provide guidelines to enhance the design of future LLM-based *resume generation* and *resume matching* systems, aiming for enhancing the future of work with AI and informing the design of human-centric AI-based tools in hiring contexts

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